Professor Q&A sheet — September 15, 2026 (Research proposal)

Retrospective reconstruction, prepared October 2026

Short answers to likely questions. "Later outcome" notes report what the completed study (Oct 6–8, 2026) found. Use them only if asked; they were not known on Sept 15.

1. Is this the deck you presented on September 15? No. It is a reconstruction prepared in October 2026. The repository record begins on Oct 6, 2026. Two sources describe the pre-October project:

- the decision record: a generic QML study on Iris, QM9 and WDBC with a mitigate-or-not decision (`docs/DECISIONS.md`, Q1);
- the early-October research plan's description of the deck of that time: a benefit-minus-cost rule, a VQC, Qiskit Aer, ZNE and readout mitigation, a Never/Always/Fixed/Q-TRACE comparison, and "current accuracy" as an example input.

The record does not say which of these were shown on Sept 15. Original September materials, if they exist, take precedence.

2. Why these benchmarks? The record names Iris and WDBC (classification) and QM9 (molecular properties) as the pre-October scope. *Later outcome:* on Oct 6 the scope was narrowed to credit-card fraud detection (decision Q1), a domain where calibrated, threshold-based decisions matter.

3. Which mitigation techniques? Zero-noise extrapolation and readout mitigation, as the research plan describes the deck, with the decision framed as mitigate or not. *Later outcome:* combined mitigation and an extra-shots action were added from Oct 6 onward.

4. Is mitigation always worth it? No. A resource-normalized study found mitigation beneficial on average but strongly platform-dependent (Russo et al., IEEE TQE 2023). That uncertainty motivates a per-context decision.

5. How would Q-TRACE decide? By weighing expected benefit against extra cost, compared with never mitigating, always mitigating and a fixed rule. *Later outcome:* in the studied setting, a simple probe rule was enough, and no learned controller was justified.

6. Will you use real quantum hardware? The record names Qiskit Aer simulation. *Later outcome:* all experiments were simulated, and hardware validation remains future work.

References: Preskill 2018 (Quantum 2:79); Temme et al. 2017 (PRL 119:180509); Kandala et al. 2019 (Nature 567:491); Giurgica-Tiron et al. 2020 (IEEE QCE); Bravyi et al. 2021 (PRA 103:042605); Russo et al. 2023 (IEEE TQE 4:1–18).